

ECON 0150 | Spring 2026 | Homework 5.1

Due: Sunday April 12, at 11:59PM

Homework is designed to both test your knowledge and challenge you to apply familiar concepts in new applications. Answer clearly and completely. You are welcomed and encouraged to work in groups so long as your work is your own. Use the provided datasets to answer the following questions. Then submit your figures and answers to Gradescope.

Q1. Revisiting the Recession and Health

In Homework 4.1, you estimated a simple relationship between county unemployment and BMI using 2011 BRFSS data, inspired by [Zhang, Lamichhane, & Wang \(2014\)](#). Your simple model was:

$$\text{BMI5}_i = \beta_0 + \beta_1 \times \text{unemployment_rate}_i + \varepsilon_i \quad (1)$$

But as Zhang et al. note, individual characteristics like gender and education may confound this relationship. In Part 5.1 we developed models with categorical controls to account for group-level differences. Let's revisit the BRFSS data and add them.

a) A researcher wants to test whether county unemployment affects BMI, while controlling for gender. Write a regression model to test this question.

b) Start with a simple fixed-effect model:

$$\text{BMI5}_i = \beta_0 + \beta_1 \times \text{Female}_i + \varepsilon_i \quad (2)$$

Report the estimated coefficients. What does $\hat{\beta}_0$ represent in context? What does $\hat{\beta}_1$ tell us about the gender gap in BMI?

c) Now fit the model you wrote in part (a):

$$\text{BMI5}_i = \beta_0 + \beta_1 \times \text{unemployment_rate}_i + \beta_2 \times \text{Female}_i + \varepsilon_i \quad (3)$$

Report the estimated coefficients. Interpret each one in context.

d) Compare the coefficient on `unemployment_rate` in part (c) to the one you estimated in HW 4.1 (without the gender control). Did it change much? What does this tell you about whether gender is a confounding variable for the unemployment–BMI relationship?

e) What does this model assume about how unemployment affects BMI for men versus women? (Hint: think about what "parallel lines" means in this context.)

Q2. Reading Regression Output

Zhang et al. (2014) progressively add controls to their models. Below is output from a similar analysis: a researcher uses data from $n = 150$ US counties to study whether hospital admission rates (per 100,000 residents) differ between high-poverty and low-poverty counties, controlling for the share of residents without health insurance. They estimate:

$$\text{admission_rate}_i = \beta_0 + \beta_1 \times \text{high_poverty}_i + \beta_2 \times \text{uninsured_pct}_i + \varepsilon_i \quad (4)$$

where $\text{high_poverty} = 1$ if the county poverty rate exceeds 15%, and 0 otherwise.

	coef	std err	t	P> t	[0.025	0.975]
Intercept	320.50	28.400	11.285	0.000	264.40	376.60
high_poverty	145.30	35.600	4.082	0.000	75.10	215.50
uninsured_pct	8.70	1.850	4.703	0.000	5.04	12.36

a) Interpret the intercept (320.50) in context.

b) Interpret the coefficient on `high_poverty` (145.30) in context.

c) Interpret the coefficient on `uninsured_pct` (8.70) in context.

d) What is the predicted admission rate for a high-poverty county where 20% of residents are uninsured?

e) What is the predicted admission rate for a low-poverty county with the same 20% uninsured rate?